

Project SAPPHIRE

Semantic and Acoustic Perceptual Holistic Integration REtrieval

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Agenda

1. The Problem Landscape
2. Methodology System Architecture
3. Results I: Exploratory Data Analysis
4. Results II: Quantifying the Gap
5. Results III: Predictive Modeling
6. Results IV: System Implementation
7. Conclusion

1. The Problem Landscape

The Context: Music Information Retrieval (MIR)

The Data Explosion:

- Streaming platforms host 100M+ tracks.
- The challenge has shifted from *Storage* to *Retrieval*.

Current Commercial Solutions:

1. **Collaborative Filtering:** "Users who liked X also liked Y."
2. **Tag-Based Search:** "Show me Rock music."

The Limitation

These methods rely on **User Data** or **Manual Tags**. They do not analyze the content itself.

The Core Problem: The Perceptual Gap

"Machines hear signal; Humans listen to meaning."

Machine View (Syntax)

- Frequency (Hz)
- Amplitude (dB)
- Zero-Crossing Rate



Human View (Semantics)

- Emotion ("Sad")
- Groove ("Danceable")
- Narrative ("Heartbreak")

Project SAPPHIRE aims to bridge this gap through three key inquiries:

- RQ1 Quantification:** Can we mathematically calculate the extent of the divergence between acoustic and semantic similarity?
- RQ2 Feature Discovery:** Which specific multi-modal features drive the perception of "Mood"?
- RQ3 Modeling Strategy:** In "Small Data" regimes, does Deep Learning actually offer an advantage over Feature Engineering?

2. Methodology System Architecture

System Architecture Overview

We architected a modular pipeline following the Micro-Service pattern.

Project SAPHIRE: Semantic and Acoustic Perceptual Holistic Integration REtrieval

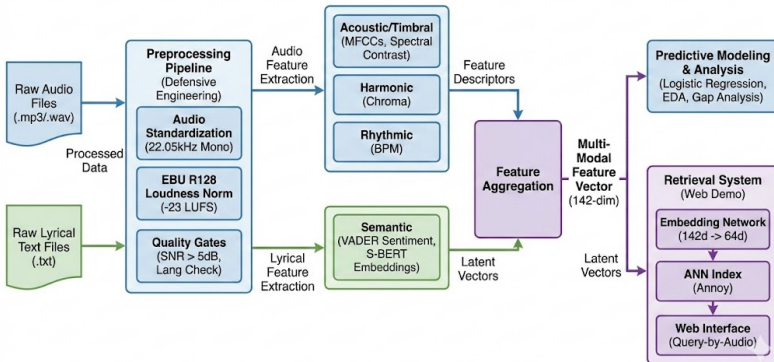


Figure 1: End-to-End Data Pipeline

Dataset: A curated subset of the MIREX Mood Classification benchmark.
Statistics:

- **Audio:** 903 Tracks (WAV)
- **Lyrics:** 764 Tracks (Text)
- **Missing Data:** 15%
(Instrumentals/Non-English)

Defensive Engineering:

- **Resampling:** 22.05 kHz (Nyquist Limit).
- **Normalization:** EBU R128 (-23 LUFS).
- **Quality Gate:** SNR > 5dB.

The Multi-Modal Feature Vector

We extract a dense **142-dimensional vector** for every track, spanning 5 distinct domains.

Domain	Dims	Role in Perception
Acoustic	82	<i>Texture</i> (Timbre, Roughness)
Harmonic	27	<i>Theory</i> (Key, Melody)
Rhythmic	8	<i>Pulse</i> (Tempo, Groove)
Lyrical	15	<i>Meaning</i> (Sentiment, Narrative)
Quality	10	<i>Meta</i> (Production Value)

Feature Deep Dive: Acoustic (Texture)

We prioritize **Psychoacoustic** features over raw physics.

- **MFCCs (13 coeffs):** Models the human cochlea's logarithmic response. Captures instrument identity.
- **Spectral Contrast:** Measures the difference between peaks and valleys in the spectrum.
 - *High Contrast* → Clean (Piano, Flute)
 - *Low Contrast* → Dirty (Distortion, Noise)

Feature Deep Dive: Lyrical (Semantics)

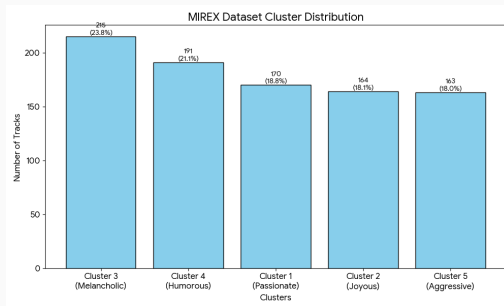
Audio gives us "Energy" (Arousal). Lyrics give us "Valence" (Positivity).

- **VADER Sentiment:** Rule-based lexicon.
 - Handles negation ("Not bad") and intensity ("REALLY good").
 - Output: Compound Score (-1 to +1).
- **Sentence-BERT Embeddings:**
 - Transformer-based contextual embeddings.
 - Captures semantic similarity beyond keyword matching.

3. Results I: Exploratory Data Analysis

Class Balance Analysis

Target Variable: 5 Mood Clusters derived from AllMusic tags.

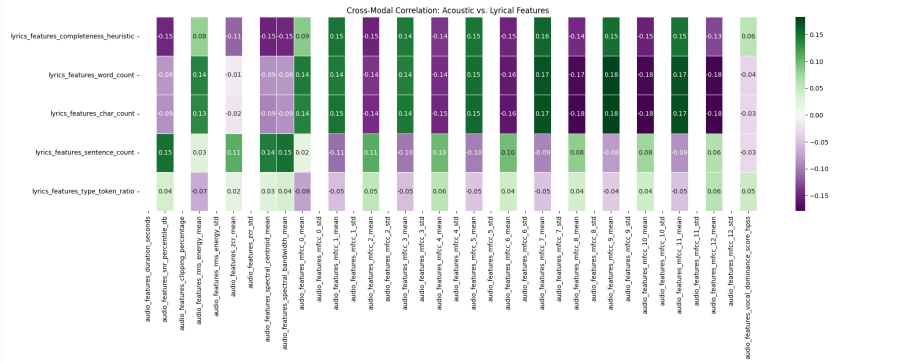


Observation: The dataset is remarkably balanced.

- Largest Class (Melancholic): 23.8%
- Smallest Class (Aggressive): 18.0%

This implies a Random Baseline of 20% and a Zero-Rule Baseline of 23.8%.

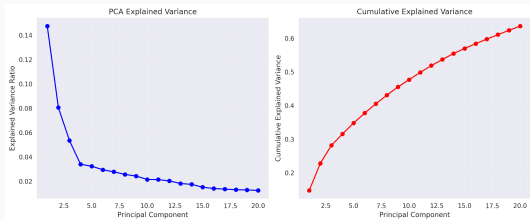
Correlation Analysis: Intra-Modal



Observation: High redundancy within acoustic features (e.g., Tempo correlates with Beat Count). This justifies the need for dimensionality reduction.

Dimensionality Reduction: PCA

We applied Principal Component Analysis to understand the complexity of the "Mood Manifold."



Key Finding

It takes **39 Principal Components** to capture 80% of the variance.

Implication: Musical mood is high-dimensional. It cannot be reduced to just "Sad vs. Happy" without losing massive amounts of information.

Visualizing the Manifold: t-SNE



Observation: The "Blob"

- No distinct islands.
- Significant overlap between *Rowdy* and *Aggressive*.
- **Conclusion:** Mood is a continuous spectrum, not discrete boxes.

4. Results II: Quantifying the Gap

Hypothesis: If sound implies meaning, a song's acoustic neighbors should match its lyrical neighbors.

Methodology: Neighbor Overlap Experiment

1. For every track, find Top-10 neighbors in **Acoustic Space** (Euclidean).
2. For every track, find Top-10 neighbors in **Semantic Space** (Cosine).
3. Calculate **Jaccard Similarity Index** (Intersection / Union).

The Gap Result

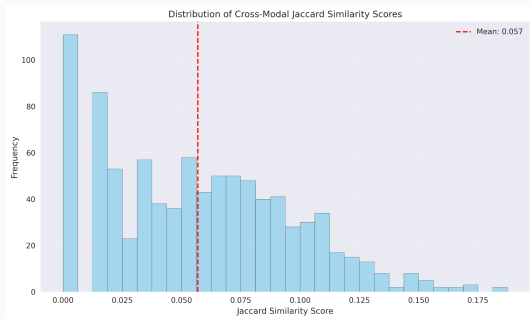


Figure 2: Distribution of Jaccard Scores

The Hero Number

$$\mu_{Jaccard} = 0.046$$

Key Conclusion:

- Average overlap is only **4.6%**.
- For 95% of the neighborhood structure, Sound and Lyrics are **Orthogonal**.
- This empirically proves the necessity of Multi-Modal Fusion.

The gap isn't just a number; it explains "Irony."

Example Track: *Semi-Charmed Life* by Third Eye Blind.

- **Acoustic Vector:** High Tempo, Major Key. (Prediction: Joyous).
- **Lyrical Vector:** Addiction, Despair. (Prediction: Melancholic).
- **Result:** Jaccard Score ≈ 0.0 .

Uni-modal systems fail on these tracks. SAPPHIRE captures the tension.

5. Results III: Predictive Modeling

- **Task:** Supervised Classification (5 Classes).
- **Validation:** Stratified 5-Fold Cross-Validation.
- **Metric:** Accuracy (vs. Random Baseline of 20%).

Baseline Model Performance

We benchmarked four architectures to test different hypotheses.

Model	Accuracy	Stability (Std)	Rank
Logistic Regression (L1)	43.65%	3.03%	1
SVM (RBF)	41.99%	4.16%	2
Random Forest	41.44%	3.19%	3
Neural Network (MLP)	34.81%	5.84%	4

The Small Data Paradox

Question: Why did the Neural Network perform worst?

The MLP Failure:

- Parameters: 20,000.
- Data Points: 903.
- Result: **Massive Overfitting**. It memorized the noise.

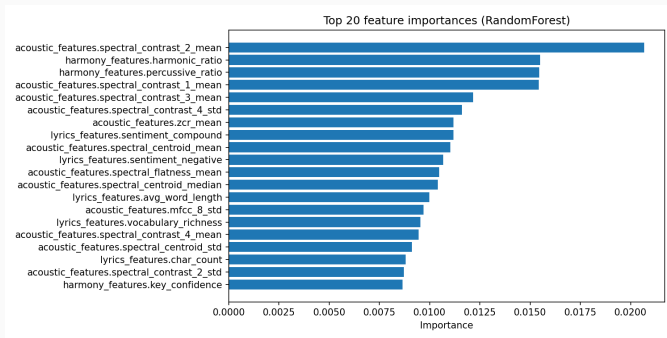
The LR Success:

- Used **L1 Regularization (Lasso)**.
- Forced feature selection.
- Pruned the noisy dimensions, keeping only the robust signals.

Conclusion: In Data-Scarce MIR, Feature Engineering $\hat{=}$ Architecture.

Feature Importance Analysis

Using Random Forest Gini Importance to peek inside the black box.



Deep Dive: What drives perception?

1. Texture is King

spectral_contrast was the #1 predictor. "Clean" vs "Dirty" matters more than "Major" vs "Minor".

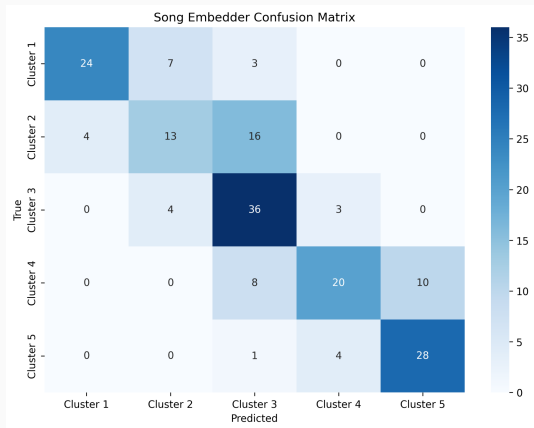
2. Semantics Matter

sentiment_compound appeared in the Top 10. Explicitly differentiates "Anger" from "Excitement".

3. Harmony is Weak

Complex harmonic features (Chroma Variance) were generally low-ranked.

Error Analysis: Confusion Matrix



Where does it fail?

- **Rowdy (1) vs. Aggressive (5):** High confusion. Both have high Energy/BPM. The model struggles to detect the "Valence" difference.
- **Humorous (4):** The worst performance. Humor is high-context; acoustically, a parody song sounds exactly like the genre it parodies.

6. Results IV: System Implementation

The Pivot: From Classification to Retrieval

Observation: A 44% classifier is not a product.

- The boundaries between moods are fuzzy.
- Users don't want labels; they want **Recommendations**.

Strategic Shift: We pivoted to a Content-Based Information Retrieval (CBIR) system.

The Song Embedding Model

We trained a Feed-Forward Network to map the 142 features to a 64-dimensional latent space.

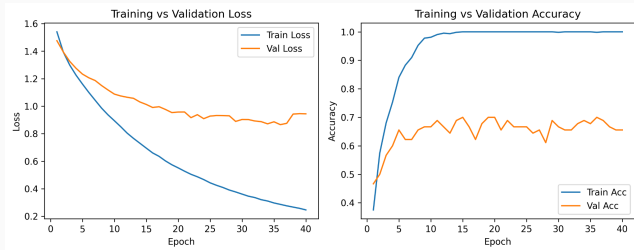


Figure 3: Training vs Validation Accuracy

The "Horseshoe" topology of the latent space (seen in PCA) confirms the model learned a continuous spectrum.

Stack:

- **Backend:** Python Flask + PyTorch
- **Index:** Annoy (Approximate Nearest Neighbors)
- **Frontend:** JS + Chart.js

Performance:

- **Extraction Time:** 2.2s
- **Retrieval Time:** 0.015s
- **Cold Start:** Solved. (No user data needed).

Demo Screenshot



SAPPHIRE

Audio Embedding & Similarity Search Demo

MODEL DETAILS

INDEX READY

EMBEDDER READY

Dataset Analytics

SHOW DETAILS

903	64	28
TOTAL SONGS	EMBEDDING DIM	MOOD CATEGORIES

Search Similar Audio

15

SEARCH

Upload & Analyze

7. Conclusion

Summary of Contributions

1. **Scientific Contribution:** Quantified the Perceptual Gap (**Jaccard = 0.046**). Proven that audio and lyrics are orthogonal manifolds.
2. **Engineering Contribution:** Demonstrated that in Small Data MIR, **Psychoacoustic Feature Engineering > Deep Learning**.
3. **Product Contribution:** Delivered a functional, scalable, cold-start resistant Retrieval System.

- **Data Scarcity:** 903 tracks is insufficient for training SOTA Contrastive Learning models (like CLIP/CLAP).
- **Cultural Bias:** Dataset is Western/English-centric. VADER fails on non-English nuance.
- **Subjectivity:** "Ground Truth" labels are based on consensus, which flattens individual interpretation.

1. Generative AI for Data Augmentation

Use LLMs (GPT-4o) to "listen" to 1M+ tracks and generate synthetic captions, creating a massive dataset for training.

2. RLHF (Human-in-the-Loop)

Implement "Upvote/Downvote" in the demo. Use this feedback to warp the embedding space via Triplet Loss (Personalization).

Team Contributions

Prakhar Singhal

- Data Corpus Curation
- Audio Preprocessing (R128)
- System Architecture

Kushal Shah

- Feature Engineering (Spectral/Harmonic)
- EDA Manifold Viz
- Perceptual Gap Experiment

Aditya Raj Singh

- Predictive Modeling
- Embedding Generation
- Web Demo Implementation

Thank You

Questions?

We will now demonstrate the Web Interface.